

KARL O. MERCADO

Software Engineer · Full-Stack & AI Developer

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PROFILE SUMMARY

UPLB CS graduate with production-deployed full-stack and AI applications. Built a RAG pipeline from scratch — Anthropic API, pgvector, Docker, CI/CD — and shipped real client work with Next.js 15 and React 19. Seeking a remote junior software engineering role where I can ship production-quality work from day one.

TECHNICAL SKILLS

Languages	JavaScript · TypeScript · Python · PHP · Java · C · Dart
Frontend	React · Next.js · Tailwind CSS · HTML5 · CSS3 · Framer Motion
Backend	Node.js · Express.js · PHP Laravel · REST APIs
Databases	PostgreSQL · MongoDB · Supabase (pgvector) · MariaDB · Firebase
AI / ML	Anthropic API · Voyage AI · RAG Pipelines · LLM API Integration · Vector Search
DevOps	Docker · GitHub Actions · CI/CD · Railway · Vercel · Ubuntu Linux
Tools	Git · GitHub · WordPress · Figma · Postman

WORK EXPERIENCE

Web Developer (Freelance) · Peaks & Pedals Transient	<i>Jan 2026 – Present</i>
- Designed and deployed a production vacation rental website achieving Lighthouse scores of 100 desktop / 95 mobile - Gathered client requirements, iterated on design based on feedback, managed domain setup and Vercel deployment independently - Built with Next.js 15, React 19, Tailwind CSS 4 — full project ownership from UI to deployment; SEO via JSON-LD, OpenGraph, and AVIF/WebP image optimization	
Software Engineer Intern · Argon Software	<i>Jun 2024 – Aug 2024</i>
- Engineered a voucher management module in PHP Laravel — handling voucher creation, user assignment, and validation logic with secure role-based access - Improved mobile responsiveness of the platform's landing page across screen sizes using CSS and Laravel Blade templates	
Web Developer (Contract) · Municipal Government Office	<i>May 2024 – Jun 2024</i>
Built and deployed a responsive WordPress government website, organizing 50+ pages of public service content with accessible navigation and official branding	

PROJECTS

MayHapotTabi — AI Document Chat Application
<i>React · TypeScript · Node.js · Express · Supabase · pgvector · Anthropic API · Docker · GitHub Actions</i>
- Engineered a full RAG pipeline from scratch: PDF ingestion → 400-token chunking → Voyage AI embeddings → pgvector cosine similarity search → Anthropic API streaming response, live on Railway + Vercel - Enforced per-user data isolation via Supabase Row Level Security — zero cross-user data leakage by design - Automated deployment with Docker multi-stage builds and GitHub Actions CI/CD, reducing manual release steps to zero
uHOME — Housing Management System with GIS
<i>React · Node.js · MongoDB · Leaflet.js · Firebase</i>
Built a GIS-enabled web app for UPLB Housing Office managing 50+ housing units with interactive map, real-time maintenance request tracking, role-based access, and secure authentication

EDUCATION

Bachelor of Science in Computer Science University of the Philippines Los Baños (UPLB)	<i>2019 – 2025</i>
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